

DreamLog

Compression Enables Generalization in Logic Programming

Alexander Towell

Southern Illinois University Edwardsville

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Three questions

- 1 Is this paper interesting enough for you to coauthor?
- 2 Does it fit into my dissertation, or is it a side project?
- 3 What would strengthening it look like before submission?

Status

- Preprint on Zenodo (DOI 10.5281/zenodo.19490027)
- 9-page draft, code and experiments public on GitHub
- Reviewed twice (self-review via a multi-agent reviewer); open issues are known

Problem. Knowledge bases grow but do not learn. You can add 300 facts about a family tree, and the system still cannot tell you whether a new person is someone's father. The facts accumulate, understanding does not.

Claim. Compression *is* learning (Solomonoff, MDL). The shortest KB consistent with the data generalizes best.

System. DreamLog, a logic programming system with wake-sleep cycles inspired by DreamCoder:

- **Wake:** answer queries via SLD resolution, optionally call an LLM to propose rules for undefined predicates.
- **Sleep:** run 8 compression operations (7 symbolic, 1 LLM-assisted), each verified against positive and negative queries with atomic rollback.

Every compression is behaviorally equivalent to the original KB on the verification suite. Nothing gets “worse.”

The result I care most about

A synthetic crafting domain. 15 materials with invented names (*lumite*, *vexal*, *drossite*, ...). The LLM has never seen these terms in training.

33 checks on unseen materials / artisans / recipes, 5 runs per condition:

Condition	Recall	Precision	Rules
No dream	0%	—	0
Raw LLM prompting	0%	—	0
Symbolic only	53%	59%	5
Full pipeline	$64 \pm 2\%$	64%	20

The raw LLM scores 0% because the terms are invented. Symbolic compression alone hits 53% without any LLM involvement, which means MDL-driven generalization discovers real concepts from data structure, not from prior knowledge. The LLM adds another 11 points on top.

This is the cleanest evidence I have for the thesis.

Strong

- Novel-domain test isolates compression from LLM memorization
- Zero false positives in either verification mode
- Open-world / closed-world distinction is a real methodological contribution
- Reproducible: all experiments scripted, registry tracks provenance
- Scales to 443 assertions, 78 predicates (research lab simulation)
- LLM cost under \$0.05 for all experiments combined

Missing

- No head-to-head ILP baseline (Popper, Metagol)
- No figures; a wake-sleep diagram would help
- Single-author empirical paper is a harder venue sell
- Recursive rule discovery fails (ancestor predicate)
- Scale is tens of predicates, not thousands

What coauthoring might look like. Where I think you could contribute:

- Statistical rigor on the empirical claims (your ransomware work set a bar for how to report these)
- Venue strategy; a senior coauthor materially changes what's reachable
- Framing with respect to my dissertation: is this a chapter or a side project?
- Advocating for a Popper benchmark run, which is mostly student-time effort but needs to be scoped

What I'd do. Everything else: bug fixes, experiments, writing, figures, the ILP benchmark setup, revisions.

- ❶ Dissertation fit. The listed advising scope is AI safety, interpretability, network analysis. DreamLog produces interpretable rules from unstructured KBs. Does that count?
- ❷ Venue. NeSy 2026 (Lisbon, September), NeurIPS workshop (December), or ILP 2027? Each has different bar, different timeline.
- ❸ Priority ordering. If strengthening, what goes first: Popper benchmark, figures, or scale?